Questions

1. Prove by induction that

$$n^5 - n$$

is divisible by 5 for all positive integers n **[5]**

Solution

We prove by induction that $n^5 - n$ is divisible by 5 for all positive integers n.
Base case: $n = 1$

$$1^5 - 1 = 1 - 1 = 0$$

Since 0 is divisible by 5, the statement is true when $n = 1$.
Inductive hypothesis:
Assume that for some positive integer k, the statement is true for $n = k$. So assume $k^5 - k$ is divisible by 5. Equivalently, there exists an integer m such that

$$k^5 - k = 5m$$

Inductive step:
We must show that $(k+1)^5 - (k+1)$ is also divisible by 5.
Let $f(n) = n^5 - n$. Then

$$f(k+1) = f(k) + \big(f(k+1) - f(k)\big)$$

Now calculate the difference:

$$\begin{aligned} f(k+1) - f(k) &= \big((k+1)^5 - (k+1)\big) - \big(k^5 - k\big) \\ &= (k+1)^5 - k^5 - 1 \\ &= \big(k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1\big) - k^5 - 1 \\ &= 5k^4 + 10k^3 + 10k^2 + 5k \\ &= 5(k^4 + 2k^3 + 2k^2 + k) \end{aligned}$$

So

$$(k+1)^5 - (k+1) = (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k)$$

By the inductive hypothesis, $k^5 - k$ is divisible by 5. Also, $5(k^4 + 2k^3 + 2k^2 + k)$ is clearly divisible by 5. Therefore $(k+1)^5 - (k+1)$ is divisible by 5.
Conclusion:
The statement is true for $n = 1$, and if it is true for $n = k$, then it is true for $n = k + 1$. Hence, by mathematical induction,

$$n^5 - n \text{ is divisible by 5 for all positive integers } n$$

2. Prove by induction that

$$9^n - 8n - 1$$

is divisible by 64 for all positive integers $n \in \mathbb{Z}^+$ **[7]**

Solution

Let

$$P(n): \quad 9^n - 8n - 1 \text{ is divisible by } 64$$

We prove $P(n)$ is true for all positive integers n by induction.
Base case: $n = 1$

$$9^1 - 8(1) - 1 = 9 - 8 - 1 = 0$$

Since 0 is divisible by 64, $P(1)$ is true.
Inductive hypothesis
Assume that $P(k)$ is true for some positive integer k.
So we assume

$$9^k - 8k - 1 \text{ is divisible by } 64$$

Hence there exists an integer m such that

$$9^k - 8k - 1 = 64m$$

Inductive step
We now consider

$$9^{k+1} - 8(k+1) - 1$$

$$\begin{aligned} 9^{k+1} - 8(k+1) - 1 &= 9 \cdot 9^k - 8k - 8 - 1 \\ &= 9 \cdot 9^k - 8k - 9 \\ &= 9(9^k - 8k - 1) + 64k \end{aligned}$$

Now use the inductive hypothesis $9^k - 8k - 1 = 64m$:

$$\begin{aligned} 9^{k+1} - 8(k+1) - 1 &= 9(64m) + 64k \\ &= 64(9m + k) \end{aligned}$$

Since $9m + k$ is an integer, $9^{k+1} - 8(k+1) - 1$ is divisible by 64.
Therefore $P(k+1)$ is true.
Conclusion
Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, it follows by induction that

$$9^n - 8n - 1 \text{ is divisible by } 64 \text{ for all positive integers } n$$

3. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$13^n + 3 \cdot 4^n + 11$$

is divisible by 12 **[7]**

Solution

(a) Let

$$P(n) : \text{“}13^n + 3 \cdot 4^n + 11 \text{ is divisible by } 12\text{”}$$

We prove $P(n)$ is true for all $n \in \mathbb{Z}^+$ by induction.

Base case: $n = 1$**.**

$$13^1 + 3 \cdot 4^1 + 11 = 13 + 12 + 11 = 36 = 12 \cdot 3$$

So $13^1 + 3 \cdot 4^1 + 11$ is divisible by 12. Hence $P(1)$ is true.

Inductive hypothesis.

Assume that $P(k)$ is true for some $k \in \mathbb{Z}^+$. Then for some integer m,

$$13^k + 3 \cdot 4^k + 11 = 12m$$

Inductive step.

We must show that $P(k+1)$ is true, i.e. that $13^{k+1} + 3 \cdot 4^{k+1} + 11$ is divisible by 12.

Consider the difference:

$$\begin{aligned}
&\left(13^{k+1} + 3 \cdot 4^{k+1} + 11\right) - \left(13^k + 3 \cdot 4^k + 11\right) \\
&= 13^{k+1} - 13^k + 3 \cdot 4^{k+1} - 3 \cdot 4^k \\
&= 13^k(13 - 1) + 3 \cdot 4^k(4 - 1) \\
&= 12 \cdot 13^k + 9 \cdot 4^k \\
&= 12 \cdot 13^k + 12\left(3 \cdot 4^{k-1}\right)
\end{aligned}$$

Since $k \geq 1$, 4^{k-1} is an integer, so this whole difference is a multiple of 12.

Therefore

$$\begin{aligned}
13^{k+1} + 3 \cdot 4^{k+1} + 11 &= \left(13^k + 3 \cdot 4^k + 11\right) + \left[\left(13^{k+1} + 3 \cdot 4^{k+1} + 11\right) - \left(13^k + 3 \cdot 4^k + 11\right)\right] \\
&= 12m + 12 \cdot 13^k + 12\left(3 \cdot 4^{k-1}\right) \\
&= 12\left(m + 13^k + 3 \cdot 4^{k-1}\right)
\end{aligned}$$

So $13^{k+1} + 3 \cdot 4^{k+1} + 11$ is divisible by 12. Hence $P(k+1)$ is true.

Since $P(1)$ is true and $P(k) \Rightarrow P(k+1)$, it follows by mathematical induction that

$$13^n + 3 \cdot 4^n + 11$$

is divisible by 12 for all $n \in \mathbb{Z}^+$.

4. Prove by mathematical induction that, for all non-negative integers n,

$$2^n + 2 \cdot 3^{2n} + 4^{2n+1}$$

is divisible by 7 **[6]**

Solution

We prove the statement by mathematical induction.
Let $P(n)$ be the statement

$$2^n + 2 \times 3^{2n} + 4^{2n+1} \text{ is divisible by } 7$$

Base case: $n = 0$**.**

$$2^0 + 2 \times 3^0 + 4^1 = 1 + 2 + 4 = 7$$

Since 7 is divisible by 7, $P(0)$ is true.
Inductive hypothesis.
Assume that $P(k)$ is true for some non-negative integer k. So

$$2^k + 2 \times 3^{2k} + 4^{2k+1}$$

is divisible by 7. Hence, for some integer m,

$$2^k + 2 \times 3^{2k} + 4^{2k+1} = 7m$$

Inductive step.
We must show that $P(k+1)$ is true, that is, that

$$2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3}$$

is divisible by 7.
Starting with this expression,

$$\begin{aligned} 2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3} &= 2 \cdot 2^k + 2 \cdot 3^2 \cdot 3^{2k} + 4^2 \cdot 4^{2k+1} \\ &= 2 \cdot 2^k + 18 \cdot 3^{2k} + 16 \cdot 4^{2k+1} \\ &= 2\big(2^k + 2 \times 3^{2k} + 4^{2k+1}\big) + 14 \cdot 3^{2k} + 14 \cdot 4^{2k+1} \end{aligned}$$

Now use the inductive hypothesis $2^k + 2 \times 3^{2k} + 4^{2k+1} = 7m$:

$$\begin{aligned} 2^{k+1} + 2 \times 3^{2k+2} + 4^{2k+3} &= 2(7m) + 14 \cdot 3^{2k} + 14 \cdot 4^{2k+1} \\ &= 14m + 14\big(3^{2k} + 4^{2k+1}\big) \\ &= 7\big(2m + 2 \cdot 3^{2k} + 2 \cdot 4^{2k+1}\big) \end{aligned}$$

This is 7 multiplied by an integer, so it is divisible by 7. Therefore $P(k+1)$ is true.
Since $P(0)$ is true and $P(k) \Rightarrow P(k+1)$, it follows by mathematical induction that

$$2^n + 2 \times 3^{2n} + 4^{2n+1}$$

is divisible by 7 for all non-negative integers n.

5. For each positive integer n, define

$$u_n = 5^{2n} - 2^n$$

Use proof by induction to show that 23 is a factor of u_n for all $n \in \mathbb{Z}^+$ **[6]**

Solution

Let $P(n)$ be the statement

$$5^{2n} - 2^n \text{ is divisible by } 23$$

We prove $P(n)$ for all $n \in \mathbb{Z}^+$ by induction.
Base case: $n = 1$.

$$u_1 = 5^2 - 2^1 = 25 - 2 = 23$$

So u_1 is divisible by 23. Hence $P(1)$ is true.
Inductive hypothesis.
Assume that $P(k)$ is true for some positive integer k. So u_k is divisible by 23, which means there is an integer m such that

$$u_k = 5^{2k} - 2^k = 23m$$

Inductive step.
Now consider u_{k+1}:

$$\begin{aligned} u_{k+1} &= 5^{2(k+1)} - 2^{k+1} \\ &= 5^{2k+2} - 2 \cdot 2^k \\ &= 25 \cdot 5^{2k} - 2 \cdot 2^k \\ &= (23 + 2)5^{2k} - 2 \cdot 2^k \\ &= 23 \cdot 5^{2k} + 2 \cdot 5^{2k} - 2 \cdot 2^k \\ &= 23 \cdot 5^{2k} + 2\left(5^{2k} - 2^k\right) \end{aligned}$$

Using the inductive hypothesis $5^{2k} - 2^k = 23m$,

$$\begin{aligned} u_{k+1} &= 23 \cdot 5^{2k} + 2(23m) \\ &= 23\left(5^{2k} + 2m\right) \end{aligned}$$

Since $5^{2k} + 2m$ is an integer, it follows that u_{k+1} is divisible by 23. So $P(k+1)$ is true.
Conclusion.
Since $P(1)$ is true, and $P(k)$ true implies $P(k+1)$ true, $P(n)$ is true for all $n \in \mathbb{Z}^+$ by mathematical induction.
Therefore, 23 is a factor of $u_n = 5^{2n} - 2^n$ for all positive integers n.

6. Prove by induction that, for $n \in \mathbb{Z}^+$,

$$7^n - 6n - 1$$

is divisible by 36 **[6]**

Solution

Let $P(n)$ be the statement

$$7^n - 6n - 1 \text{ is divisible by } 36$$

We prove $P(n)$ is true for all $n \in \mathbb{Z}^+$ by mathematical induction.
Base case: $n = 1$.

$$7^1 - 6(1) - 1 = 7 - 6 - 1 = 0$$

Since $0 = 36 \times 0$, 0 is divisible by 36. So $P(1)$ is true.
Inductive hypothesis: Assume that $P(k)$ is true for some $k \in \mathbb{Z}^+$.
So we assume

$$7^k - 6k - 1$$

is divisible by 36, which means there exists an integer m such that

$$7^k - 6k - 1 = 36m$$

Inductive step: We must show that $P(k+1)$ is true, that is, that

$$7^{k+1} - 6(k+1) - 1$$

is divisible by 36.
Starting with this expression,

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7^{k+1} - 6k - 6 - 1 \\ &= 7^{k+1} - 6k - 7 \end{aligned}$$

Now rewrite it to use the inductive hypothesis:

$$\begin{aligned} 7^{k+1} - 6k - 7 &= 7 \cdot 7^k - 6k - 7 \\ &= 7(7^k - 6k - 1) + 36k \end{aligned}$$

Using the inductive hypothesis $7^k - 6k - 1 = 36m$,

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7(36m) + 36k \\ &= 36(7m + k) \end{aligned}$$

Since $7m + k \in \mathbb{Z}$, this is a multiple of 36. Therefore

$$7^{k+1} - 6(k+1) - 1$$

is divisible by 36.
So $P(k+1)$ is true.
Since $P(1)$ is true, and $P(k) \Rightarrow P(k+1)$, it follows by induction that

$$7^n - 6n - 1 \text{ is divisible by } 36 \qquad \text{for all } n \in \mathbb{Z}^+$$

Hence $7^n - 6n - 1$ is divisible by 36 for all positive integers n.

7. The function f is defined by

$$f(n) = 9^n - 8n + 63$$

Prove by induction that $f(n)$ is divisible by 64 for $n \geq 1$ **[6]**

Solution

We prove by induction that

$$9^n - 8n + 63 \text{ is divisible by } 64$$

for all $n \geq 1$.

Base case: $n = 1$.

$$f(1) = 9^1 - 8(1) + 63 = 9 - 8 + 63 = 64$$

Since 64 is divisible by 64, the statement is true for $n = 1$.

Induction hypothesis:

Assume that for some integer $k \geq 1$,

$$f(k) = 9^k - 8k + 63$$

is divisible by 64.

So we may write

$$9^k - 8k + 63 = 64m$$

for some integer m.

Induction step:

Now consider $f(k+1)$:

$$\begin{aligned} f(k+1) &= 9^{k+1} - 8(k+1) + 63 \\ &= 9 \cdot 9^k - 8k - 8 + 63 \\ &= 9 \cdot 9^k - 8k + 55 \end{aligned}$$

We now rewrite this in terms of $f(k) = 9^k - 8k + 63$:

$$f(k+1) = 9(9^k - 8k + 63) + 64(k - 8)$$

To check this expansion:

$$\begin{aligned} 9(9^k - 8k + 63) + 64(k - 8) &= 9^{k+1} - 72k + 567 + 64k - 512 \\ &= 9^{k+1} - 8k + 55 \\ &= f(k+1) \end{aligned}$$

Using the induction hypothesis $9^k - 8k + 63 = 64m$,

$$\begin{aligned} f(k+1) &= 9(64m) + 64(k - 8) \\ &= 64(9m + k - 8) \end{aligned}$$

Since m and k are integers, $9m + k - 8$ is an integer, so $f(k+1)$ is divisible by 64.

Therefore, if the statement is true for $n = k$, it is also true for $n = k + 1$.

Conclusion:

The statement is true for $n = 1$, and true for $n = k + 1$ whenever it is true for $n = k$.

Hence, by mathematical induction,

$$9^n - 8n + 63 \text{ is divisible by } 64 \quad \text{for all } n \geq 1$$

So $f(n)$ is divisible by 64 for all $n \geq 1$.

8. Prove by mathematical induction that

$$7^{2n} + 8 \times 13^n - 9$$

is divisible by 36 for all positive integers n **[6]**

Solution

We prove the statement by mathematical induction.
Let

$$E_n = 7^{2n} + 8 \times 13^n - 9$$

We want to show that E_n is divisible by 36 for all positive integers n.
Base case: $n = 1$.

$$\begin{aligned} E_1 &= 7^2 + 8 \times 13 - 9 \\ &= 49 + 104 - 9 \\ &= 144 \\ &= 36 \times 4 \end{aligned}$$

So E_1 is divisible by 36.
Inductive hypothesis: assume that for some positive integer k,

$$E_k = 7^{2k} + 8 \times 13^k - 9$$

is divisible by 36.
So there exists an integer q such that

$$7^{2k} + 8 \times 13^k - 9 = 36q$$

Inductive step: consider E_{k+1}.

$$\begin{aligned} E_{k+1} &= 7^{2(k+1)} + 8 \times 13^{k+1} - 9 \\ &= 7^{2k+2} + 8 \times 13^{k+1} - 9 \\ &= 49 \cdot 7^{2k} + 104 \cdot 13^k - 9 \end{aligned}$$

Now rewrite this so that E_k appears:

$$\begin{aligned} E_{k+1} &= 13 \cdot 7^{2k} + 36 \cdot 7^{2k} + 13 \cdot 8 \cdot 13^k - 117 + 108 \\ &= 13\left(7^{2k} + 8 \times 13^k - 9\right) + 36\left(7^{2k} + 3\right) \\ &= 13E_k + 36\left(7^{2k} + 3\right) \end{aligned}$$

Using the inductive hypothesis $E_k = 36q$,

$$\begin{aligned} E_{k+1} &= 13(36q) + 36\left(7^{2k} + 3\right) \\ &= 36\left(13q + 7^{2k} + 3\right) \end{aligned}$$

Since $13q + 7^{2k} + 3$ is an integer, E_{k+1} is divisible by 36.
So, if the statement is true for $n = k$, it is also true for $n = k + 1$.
Since the statement is true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, it follows by mathematical induction that

$$7^{2n} + 8 \times 13^n - 9$$

is divisible by 36 for all positive integers n.